AI 4092

Final Year Project

Physical Layer Security in 5G Communication

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Project Progress Report

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1. INTRODUCTION

1.1 BACKGROUND AND PROBLEM STATEMENT

The unprecedented number of connected devices in the 5G era has squeezed available radio spectrum, forcing dynamic spectrum access via cognitive radio networks. Although CRNs optimize bandwidth by allowing unlicensed Secondary Users to squander spectrum belonging to Primary Users, this shared ad hoc environment creates critical Physical Layer Security vulnerabilities. In particular, such networks are prone to malicious Radio Frequency jamming that can destroy signal integrity and deny service. To add to this difficulty, existing academic simulators that abstract signal energy rather than modeling it directly from modulated data are prone to fidelity gaps. This project solves these issues by developing a high fidelity, data-driven 5G CRN simulation framework for autonomous, real-time jamming detection and active mitigation.

1.2 ARCHITECTURAL ADVANCEMENTS AND NETWORK SCALING

Initially, we developed a Python-based client-server transceiver architecture with adaptive QAM (16/64/256) and OFDM modulation secured internally by AES-256-GCM and Reed-Solomon error correction. During the current reporting period, the core physical and medium access control layers have undergone significant architectural scaling. A dedicated spectrum sensing agent was incorporated to manage energy footprints, while strict primary/secondary pre-emption conditions were resolved for prioritized communication within the receiver logic. Additionally, the network topology was changed from a localized point-to-point link to a multi-node communication pipeline capable of handling increased bandwidth as a realistic testbed for adversarial scenarios.

1.3 ADVERSARIAL MODELING AND MACHINE LEARNING INTEGRATION

Its defining step is the successful implementation of an Artificial Intelligence driven security loop. A dynamic jamming agent was configured to inject mathematical adversarial models like Gaussian Barrage and Reactive Bit-Flipping into the shared spectrum to simulate realistic threat vectors. The project has moved away from static, rule-based thresholds towards advanced machine learning approaches to detect and counter such injected attacks. We implemented an out-of-core GPU streaming pipeline bypassing hardware constraints using a massive RF jamming dataset. Importantly, the detection pipeline has been trained & tested across five different machine learning models. This includes traditional ensemble algorithms (Random Forest, XGBoost) and advanced Deep Learning architectures (DNN, 1D-CNN, and LSTM) to classify incoming spectral signatures as clean, Gaussian, or pulsed jamming.

1.4 ACTIVE MITIGATION AND REPORT OBJECTIVES

In combination with the ML detection layer, the transceivers now actively trigger real-time mitigation protocols such as adaptive modulation and coding, as well as time-domain backoffs to avoid detected interference. We have added a set of new analytical plots and visualization tools to the simulation base station to empirically validate these mechanisms. This progress report details upgrades of the multi-node PHY/MAC architecture, comparative evaluation of five trained ML models, and demonstrates that the framework can ensure secure, high-throughput coexistence in software-defined 5G cognitive networks.

2. TIMELINE

Version	Date	Description of change
2.1 Network Scaling & Preemption	15th Jan'26	Enabled multi-node communication, implemented thread-safe architecture, and fixed primary/secondary preemption logic.
2.2 Threat Modeling & Jammer Agent	5th Feb'26	Configured malicious jamming agent, implemented air-interface packet corruption, and receiver-side anomaly detection.
2.3 Advanced Visual Analytics	25th Feb'26	Integrated headless plotting engine (Agg) and comprehensive PDF analytics (heatmaps, SINR degradation, constellation distortion).
2.4 Synthetic Dataset Generation	10th Mar'26	Implemented CTGAN methodology for stratified data loading and balanced synthetic RF data generation (TSTR evaluated).
2.5 ML Pipeline & Model Training	28th Mar'26	Trained and evaluated 5 distinct ML models (Random Forest, XGBoost, DNN, CNN1D, LSTM) on the combined dataset.

3. PROGRESS

Phase 2 saw the project move from an OFDM foundation to a multi-node Physical Layer Security testbed. Network architecture was scaled up for concurrent thread-safe transceiver communications, while primary user preemption over secondary users at the MAC layer was enforced strictly. During cyclical attack and cooldown phases, a dynamic jamming agent was integrated to model intermittent, high-power broadband interference in the shared spectrum. A Spectrum Sensing Agent was implemented to empirically analyze these network dynamics by calculating real-time energy envelopes from OFDM I/Q samples using Urkowitz detection models and a headless analytics engine exporting 3D constellation distortion and SINR degradation visualizations.

We fed an enormous 96,090-file RF Jamming dataset into active and passive spectral scans and built a tiered, hybrid training architecture to bypass hardware memory limits and power the AI-driven mitigation pipeline. Classical ensemble algorithms were trained on optimized subsets: Random Forest trained with 10,000 files in memory, and XGBoost trained with out-of-core GPU streaming for 30,000 files. In turn, our deep learning architectures (1D-CNN and LSTM) use optimized, buffered input data generators to train on the entire 96,090-file dataset. Last but not least, a Conditional Tabular GAN (CTGAN) was applied to synthesize augmenting RF datasets whose physical-layer fidelity was validated via Train-on-Synthetic, Test-on-Real metrics and Kernel Density Estimation plots.

3.1. Milestone 1: Network Scaling & Core Logic Refinement

- Enabled Multi-Node Communication: Upgraded the pipeline to handle multiple sender-receiver pairs concurrently with increased bandwidth capacities.
- Fixed Primary/Secondary Pre-emption: Correction of MAC-layer logic so that if a Secondary User sends a message, a Primary User (e.g. JAZZ, UFONE) can preempt the session and get the receiver to drop the secondary connection.
- Thread-Safe Architecture: Handling of sockets and logging with threading.Lock() refactored to prevent race conditions on high-bandwidth, multi-node communication.

3.2. MILESTONE 2: THREAT MODELING & JAMMING INFRASTRUCTURE

- Developed the Malicious Jamming Agent: Developed a high-power barrage jammer node to flood the Base Station with wideband noise to induce interference states.
- Air-Interface Corruption: Upgraded the Base Station as the physical medium. Once active, the BS performs bit-flipping (corrupting 30% of the payload) on legitimate packets to simulate real RF interference.
- Receiver-Side Anomaly Detection: Added receiver update to catch AES-GCM tag verification and Reed-Solomon decoding failures and flagged them as "JAMMED" packets.

3.3. MILESTONE 3: ADVANCED SPECTRUM SENSING & VISUAL ANALYTICS

- Integrated Spectrum Sensing Agent: Special logic was added to calculate real-time energy envelopes using Urkowitz energy detection models directly from the I/Q samples of the OFDM signals.
- Headless Plotting Engine: Use matplotlib.use('Agg') to avoid the simulation crashing under the load of creating massive data visualizations.
- Comprehensive Data Export: An automated analytics engine that exports 10 new granular plots per session.

3.4. MILESTONE 4: MACHINE LEARNING INTEGRATION

- Dataset Generation: Stratified data loading and SDV metadata definitions implemented for training conditional tabular GAN on multi-modal RF data. Exceptional balanced datasets for minority classes.
- Train 5 Different ML Models: So far, we processed the Kaggle RF Jamming dataset and your synthetic data with random forest, xgboost, deep neural network, 1D convolutional neural network, and long short-term memory.
- Final Real-Time AI Deployment Locked into the Hybrid Architecture (10,000 files for RF, 30,000 streamed files for XGBoost, and 100k files for DL models) prevents Kaggle OOM errors and allows the model to be injected directly into the Python receiver for real-time jamming identification.

4. UPDATED TIMELINE

Step	Date	Description
1.1 Initial pipeline	21st Sept'25	Demo OFDM pipeline for Proposal Defense
1.2 Structured two-way functional pipeline	10th Oct'25	Sender Receiver instance implementation along with base stations
1.3 Setting up Agent 1	20th Oct'25	Implemented Agent for Resource Allocation
1.4 Setting up Agent 2	28th Oct'25	Implement Agent For subcarrier selection
1.5 Visual representations	15th Nov'25	Fixed and added all the graphical plots
1.6 Spectrum Sensing	28th Nov'25	Introduce and implemented the spectrum sensing
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5. REFERENCES

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